

Pinching-Antenna Systems For Indoor Immersive Communications: A 3D-Modeling Based Performance Analysis

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Abstract—The emerging pinching antenna (PA) technology has high flexibility to reconfigure wireless channels and combat line-of-sight blockage, thus holding transformative potential for indoor immersive applications in 6G. This paper investigates Pinching-antenna systems (PASS) for indoor immersive communications. We construct a 3D spatial model to characterize the distributions of users, waveguides, and PAs, and develop a general theoretical framework for analyzing PASS downlink successful transmission probability by capturing random user interference and spatial coupling effects between PAs and users, as well as critical system parameters' impacts. Comprehensive numerical evaluations validate the analysis and provide practical guidelines for optimizing PASS deployments.

Index Terms—Pinching-antenna systems (PASS), indoor immersive communication, successful transmission probability.

I. INTRODUCTION

THE upcoming 6G networks unlock a new era of immersive communications by seamlessly integrating physical infrastructures and virtual worlds, facilitating the widespread adoption of extended reality, metaverse, and digital twin [1], [2]. As shown in Fig. 1, benefited by immersive communications, a variety of indoor applications are emerging, such as holographic displays, virtual tourism, online gaming, and intelligent manufacturing. These indoor immersive applications will fundamentally revolutionize the way people work, entertain, and communicate through lifelike interactions and immersive experiences. However, the devices used for these applications need to operate at high-frequency bands to achieve high-speed and low-latency transmissions, which makes them

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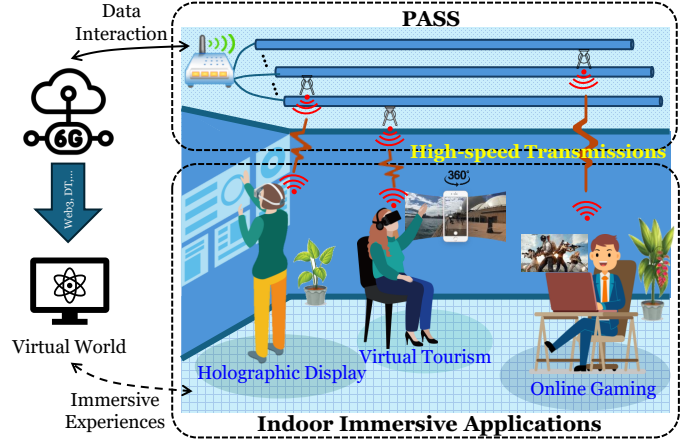


Fig. 1: PASS for indoor immersive communications.

highly susceptible to severe path loss and LoS blockage caused by walls, furniture, and human movements [3]. Consequently, these applications impose stringent requirements on network capacity and reliability while facing serious communication vulnerabilities in fixed-position conventional wireless systems, especially under dense user access, random blockages, and strong multipath propagation in immersive environments.

To address these issues, flexible and reconfigurable wireless technologies are required to enhance link reliability and network adaptability for indoor immersive applications. Pinching antennas (PAs), first demonstrated by NTT DOCOMO in 2021 [4], represent a groundbreaking technology that utilizes dielectric waveguides to guide electromagnetic (EM) waves at high frequencies. When a small dielectric element, such as a plastic clothespin, is attached to the waveguide, EM coupling occurs, enabling controllable radiation and reconfigurable signal paths [5], [6]. Building upon this mechanism, the pinching-antenna system (PASS) enables flexible PA deployment along line-shaped waveguides, allowing adaptive control and positioning to dynamically establish LoS links and overcome transmission challenges from indoor blockages. An example of indoor PASS deployment is illustrated in Fig. 1. Prior studies have explored several PA applications, including PA-assisted integrated sensing and communications (ISAC) [7], [8], non-orthogonal multiple access (NOMA) [9], and simultaneous wireless information and power transfer (SWIPT) [10]. Furthermore, [3] investigated PA-assisted immersive extended reality and metaverse applications, confirming the potential of PASS in supporting high-capacity immersive communications. However, their analysis was limited to a single waveguide, PA,

and user, without considering random interference and spatial coupling effects in realistic multi-waveguide, multi-user scenarios. This limitation motivates the development of a general and accurate theoretical model for analyzing and optimizing PASS performance in indoor immersive environments.

This paper investigates downlink transmission performance for PASS-enabled indoor immersive communications. The key contributions are threefold: **First**, we construct a 3D spatial model for PASS in indoor scenarios, where users (i.e., diverse devices and sensors for immersive applications) are uniformly distributed in the serving area, each served by the nearest PA along a ceiling-mounted waveguide. **Second**, we develop a general theoretical framework for analyzing downlink performance, i.e., successful transmission probability (STP), which (i) captures the random user interference, and the spatial coupling effects between multiple PAs and users, (ii) incorporates critical system parameters such as deployment configurations and transmission settings, and (iii) enables flexible performance evaluation for users located at arbitrary positions in practical immersive environments. **Third**, we conduct comprehensive numerical evaluations to validate the theoretical analysis and provide practical guidelines for optimizing PASS design to enhance the performance of PA-assisted indoor immersive communications.

II. SYSTEM MODEL

As shown in Fig. 2(a), an access point (AP) equipped with a PASS serves users uniformly distributed in a service area, e.g., a room. Multiple dielectric waveguides of PASS are evenly installed on the ceiling to cover the whole room. Each waveguide is equipped with an adjustable PA controlled by the AP. The AP facilitates downlink transmissions for users with immersive experiences.

A. Deployment Model

The service area is approximated as a rectangular cuboid with length L , width D , and height h . In the PASS, $2K + 1$ waveguides are installed along the wider side of the ceiling, with an inter-waveguide spacing of d among each other; and each waveguide has a length equal to the room length L . Let $\mathbf{w} = \{w_k\}_{k \in \mathcal{K}}$ denote the set of waveguides, where $\mathcal{K} = \{0, \pm 1, \dots, \pm K\}$. The central waveguide (w_0) is positioned along the midline of the ceiling's wide side, while the remaining waveguides ($w_k, k \neq 0$) are symmetrically arranged on either side of w_0 . Each waveguide w_k is connected to the AP via a dedicated radio frequency (RF) chain, with signal injection occurring at its feed point f_k . The users are uniformly distributed across the ground. Each user is served by its nearest waveguide. For a user u_k served by the waveguide w_k , the position of the PA p_k is the nearest location of its user, i.e., the top of the user. Considering the inter-waveguide spacing d , the service regions of the waveguides on the ground can be modeled as multiple rectangular areas, each with a length of L and a width of d , directly beneath the corresponding waveguides. It is worth noting that $(2K + 1)d \leq D$ is required to ensure the deployment of waveguides in the room.

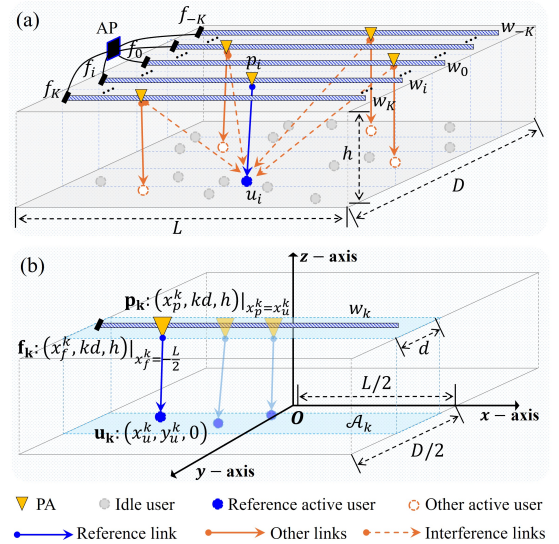


Fig. 2: System model, where (a) shows the deployment of the PASS in a room and (b) shows the used 3D coordinate system, where the blue rectangle area represents the serving area of focused w_k .

B. 3D Coordinate System

We establish a 3D Cartesian coordinate system as illustrated in Fig. 2(b). The origin, denoted as O , is located at the midpoint of the room's floor. The xy -plane is on the ground, the y -axis points toward the midpoint of the floor's wider side, the x -axis is perpendicular to the y -axis and oriented in the direction opposite to the feed points of the waveguides, and the z -axis extends vertically toward the midpoint of the ceiling. Therefore, the position of every node in the PASS can be represented as a 3D coordinate (x, y, z) . For a waveguide w_k positioned at $y = kd$ and $z = h$ (i.e., on the ceiling), let f_k , p_k , and u_k denote the coordinates of its feed point f_k , its PA p_k , and its served user u_k , respectively. We have $f_k : (x_f^k, kd, h)$, $p_k : (x_p^k, kd, h)$, $u_k : (x_u^k, y_u^k, 0)$, where $x_f^k = -L/2$. Since p_k is movable along the waveguide w_k , its x -axis location has the value range $x_p^k \in [-L/2, L/2]$. Let A_k be the service region of w_k on the ground, the points on A_k has the x -axis/ y -axis values $x \in [-L/2, L/2]$ and $y \in [kd - d/2, kd + d/2]$. Given the fixed u_k , p_k is adjusted to the nearest location of the served user, i.e., $x_p^k = x_u^k$.

C. Channel Model

To provide downlink services for users, the AP passes signals into multiple waveguides. Each downlink signal propagates through the waveguide from the feed point to the PA. The PA then radiates the signal to its served user. The signal propagated through the waveguide w_k experiences a phase shift [5]. Let h_1^k be the phase shift gain between the feed point f_k and the PA p_k , which is given by $h_1^k = e^{-j \frac{2\pi}{\lambda_g} \|f_k - p_k\|}$, where j is the imaginary unit, $\lambda_g = \lambda/n_{\text{eff}}$ is the guided wavelength with n_{eff} being the effective refractive index of the dielectric waveguide [5], λ is the wavelength of the signal, and $\|f_k - p_k\|$ is the distance between f_k and p_k .

Since the PA in each waveguide is usually activated to the top of its user, the link between them is assumed as the LoS path, and the free-space channel model is used to characterize the signal propagation. Let h_2 be the channel between p_k and

its serving user u_k , which is given by $h_2^k = \frac{\sqrt{\eta}e^{-j\frac{2\pi}{\lambda}\|\mathbf{p}_k - \mathbf{u}_k\|}}{\|\mathbf{p}_k - \mathbf{u}_k\|}$, where $\eta = \lambda^2/(16\pi^2)$ is the path loss at a reference distance of 1 m, and $\|\mathbf{p}_k - \mathbf{u}_k\|$ is the distance between p_k and u_k . Recall that p_k and u_k have the same x -axis location for the nearest association. Then their distance $\|\mathbf{p}_k - \mathbf{u}_k\|$ only depends on the height of p_k and their y -axis locations.

III. PERFORMANCE MODELING

This section first provides the link model involving the transmitted signal and the received signal-to-interference-plus-noise-ratio (SINR). Accordingly, we develop the theoretical model for link performance and provide some analysis.

A. Link Model

As shown in Fig. 2(a), we consider a typical PA-assisted downlink transmission scenario, where all waveguides simultaneously work and each serves one user. The PA in each waveguide is assumed to use the identical power, i.e., $P_t = P_{\text{total}}/(2K+1)$, where P_{total} is the total radiation power of the AP [6]. Let s_w^k be the normalized signal fed into the waveguide w_k (with the PA p_k), where $k \in \mathcal{K}$.

The practical resources of spectrum are limited to support the increasing number of users. Thus, interference occurs when multiple PA-assisted downlink links compete and use the same spectrum band. For the reference link from the PA p_i to the user u_i , let s_u^i be the received signal by u_i , which is given by

$$s_u^i = \sqrt{P_t}h_1^i h_2^i s_w^i + I + w, \quad (1)$$

where $I = \sum_{k \in \mathcal{K}/\{i\}} \alpha \sqrt{P_t}h_1^k h_2^k s_w^k$ is the interference causing by other PAs ($p_k, k \in \mathcal{K}/\{i\}$) with α is the interference intensity factor to accommodate these more complex scenario, e.g., various multiple access schemes and changing transmission requirements from users. The term w denotes the additive white Gaussian noise with zero mean and variance σ^2 [5].

Let γ_u^i be the SINR measured at u_i , which is given by $\gamma_u^i = P_u^i/(I_u^i + \sigma^2)$, where P_u^i is the received power of the signal from p_i and I_u^i is the received power of aggregative interference signals from p_k , which are calculated by

$$P_u^i = P_t |h_1^i h_2^i s_w^i|^2 = \frac{P_t}{\|\mathbf{p}_i - \mathbf{u}_i\|^2} \left| e^{-2\pi j \left(\frac{\|\mathbf{r}_0 - \mathbf{p}_i\|}{\lambda_g} + \frac{\|\mathbf{p}_i - \mathbf{u}_i\|}{\lambda} \right)} \right|^2 \quad (2)$$

$$\stackrel{(a)}{=} \frac{\eta P_t}{\|\mathbf{p}_i - \mathbf{u}_i\|^2},$$

$$I_u^i = \sum_{k \in \mathcal{K}/\{i\}} \eta P_t |h_1^k h_2^k s_w^k|^2 = \sum_{k \in \mathcal{K}/\{i\}} \frac{\eta P_t}{\|\mathbf{p}_k - \mathbf{u}_i\|^2}, \quad (3)$$

where Eq. (2)(a) is resulted from $|e^{-jx}| = 1$ [3]. Referring to Eqs. (2) and (3), γ_u^i can be calculated by

$$\gamma_u^i = \frac{\|\mathbf{p}_i - \mathbf{u}_i\|^{-2}}{\eta \sum_{k \in \mathcal{K}/\{i\}} \|\mathbf{p}_k - \mathbf{u}_i\|^{-2} + \frac{\sigma^2}{P_t}} = \frac{((x_p^i - x_u^i)^2 + (y_p^i - y_u^i)^2 + h^2)^{-1}}{\eta \sum_{k \in \mathcal{K}/\{i\}} ((x_p^k - x_u^i)^2 + (y_p^k - y_u^i)^2 + h^2)^{-1} + \frac{\sigma^2}{P_t}}. \quad (4)$$

B. Transmission Performance

Given a wireless link, the signal can be successfully received and decoded only when the received SINR at the receiver exceeds a given threshold. For the reference link from the PA p_i to the user u_i , let γ_0 be the SINR threshold at u_i . The *successful transmission probability (STP)* of the reference link,

denoted by $\mathcal{P}_s$, is given by $\mathbb{P}(\gamma_u^i > \gamma_0)$. Referring to Eqs. (2) to (4), $\mathcal{P}_s$ can be evaluated by Proposition 1.

Proposition 1. Given $R = \sum_{k \in \mathcal{K}/\{i\}} r_k$ and $r_k = ((x_u^k - x_u^i)^2 + (kd - y_u^i)^2 + h^2)^{-1}$, the STP $\mathcal{P}_s$ can be evaluated by the Cumulative Distribution Function (CDF) of R , i.e.,

$$\mathcal{P}_s = F_R(z), \text{ where } z = \frac{((id - y_u^i)^2 + h^2)^{-1}}{\gamma_0} - \frac{\sigma^2}{P_t}.$$

Proof. The proof is provided in Appendix A. ■

Referring to Proposition 1, the calculation of $\mathcal{P}_s$ needs to further find $F_R(z)$. Recall that each PA's location is align with its served user and all users are independently distributed, and hence $F_R(z)$ can be derived accordingly. Thus, we have the theoretical model of $\mathcal{P}_s$ as given in Theorem 1.

Theorem 1. The STP $\mathcal{P}_s$ of the link from p_i to u_i is given by

$$\mathcal{P}_s = \frac{1}{2} - \frac{1}{\pi} \left(\frac{1}{L} \right)^{2K} \int_0^\infty \text{Im} \left(\underbrace{\int_{-\frac{L}{2}}^{\frac{L}{2}} \cdots \int_{-\frac{L}{2}}^{\frac{L}{2}}}_{2K} e^{jtA} \underbrace{\mathbf{d} \dots \mathbf{d} x_u^k}_{k \in \mathcal{K}/\{i\}} \right) \frac{dt}{t},$$

where $A = \frac{\sigma^2}{\eta P_t} + \sum_{k \in \mathcal{K}/\{i\}} (\hat{r}_k)^{-1} - (\eta \gamma_0 \hat{r}_0)^{-1}$, $\hat{r}_0 = (id - y_u^i)^2 + h^2$, $\hat{r}_k = (x_u^k - x_u^i)^2 + (kd - y_u^i)^2 + h^2$, $\text{Im}(\cdot)$ is the imaginary part.

Proof. The proof is provided in Appendix B. ■

IV. EXPERIMENTAL RESULTS

This section presents comprehensive numerical results for the proposed theoretical model of $\mathcal{P}_s$ in Section III-B. In our result, the carrier frequency $f_c = 28$ GHz, the waveguide's refractive index $n_{\text{eff}} = 1.4$ [5], and the room width $D = 66$ m are used. We consider the efficient coverage strategy $(2K+1)d = D$ and the equal power allocation strategy $P_t = P_{\text{total}}/(2K+1)$ with K and D set in the particular result. Accordingly, we analyze the impacts of the following system parameters, including the reference user u_i 's location in the whole room, the PASS settings (i.e., K , h , L , and D), and transmission configurations (i.e., P_{total} , σ^2 , and γ_0) on $\mathcal{P}_s$. Considering the uniformly distributed users in the room, 10^4 Monte Carlo simulations are implemented in MATLAB to calculate the average STP $\mathcal{P}_s$ and verify theoretical results. To establish a performance benchmark, we introduce a baseline scheme where all PAs are fixed at $x_p^k = 0$ [3], emulating a **conventional fixed-position antenna system**. In all figures, we adopt the labels ‘‘The.’’ and ‘‘Sim.’’ to differentiate the theoretical and simulation results, respectively.

In Fig. 3, we analyze the spatial distribution of $\mathcal{P}_s$ based on the reference user's location, i.e., $\mathbf{u}_i : (x_u^i, y_u^i, 0)$, in the whole room. Fig. 3(a) plots the heatmap of the theoretical result of $\mathcal{P}_s$, which exhibits two symmetric distribution patterns.

- *Pattern 1:* Given $\mathbf{u}_i$ distributed in the service area $\mathcal{A}_i : ([-L/2, L/2], [id - d/2, id + d/2], 0)$ of the waveguide w_i , $\mathcal{P}_s$'s distribution is symmetric along w_i (located at $([-L/2, L/2], id, 0)$). Particularly, $\mathcal{P}_s$ has the minimum value along w_i and symmetrically increases by moving $\mathbf{u}_i$ away from w_i along its two sides.

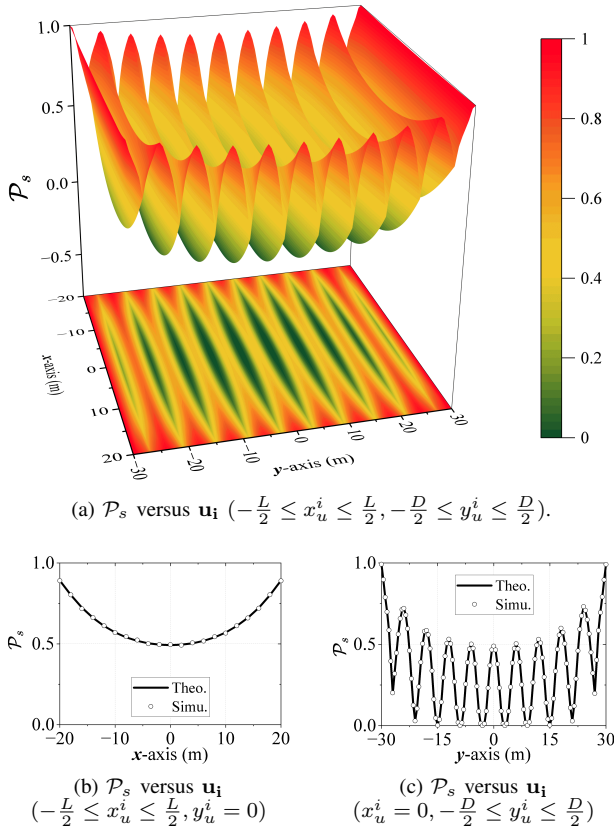


Fig. 3: P_s versus $\mathbf{u}_i : (x_u^i, y_u^i, 0)$ ($2K+1=11$, $h=3$ m, $L=40$ m, $P_{\text{total}}=1$ W, $\sigma^2=-100$ dBm, $\gamma_0=2$ dB).

- **Pattern 2:** Given $\mathbf{u}_i$ distributed in the whole room ($[-L/2, L/2], [D/2, D/2], 0$), P_s 's distribution is symmetric along the center waveguide w_0 (located at $([-L/2, L/2], 0, 0)$). Particularly, P_s periodically varies with the changing $\mathbf{u}_i$ under different service areas $\mathcal{A}_i$. The overall trend is similar to *Pattern 1*, where P_s has the minimum value along w_0 and symmetrically increases with the changing $\mathbf{u}_i$ far away from w_0 .

The above phenomena result from the systematic spatial distribution of interference across the service area of each waveguide. Undoubtedly, the user at the center of the service area (i.e., room) suffers the maximum interference by uniformly aggregating all interference signals. Meanwhile, the user served by different waveguides experiences the periodical distribution in the whole room because of the parallel distribution of waveguides. Additionally, Fig. 3(b) and Fig. 3(c) show the details of the distribution of P_s under the changing $\mathbf{u}_i$ along the x -axis and the y -axis, respectively. It can be seen that P_s increases when either values of $\|x_u^i\|$ or $\|y_u^i\|$ far away from the center 0 or id . It is because the propagation distance of the desired signal is nearly optimized by the flexible PA's location aligning with the user, thus, the longer propagation distance of the interference signal potentially enhances the SINR.

In Fig. 4, we analyze the impacts of K , h , L , D , γ_0 , P_{total} , and σ^2 on P_s . We have the following observations.

- P_s decreases as $2K+1$ increases. This is because we adopt a coverage strategy where $(2K+1)d = D$ and an equal power allocation strategy where $P_t = P_{\text{total}}/(2K+1)$, with P_{total} and D being constant. As $2K+1$ in-

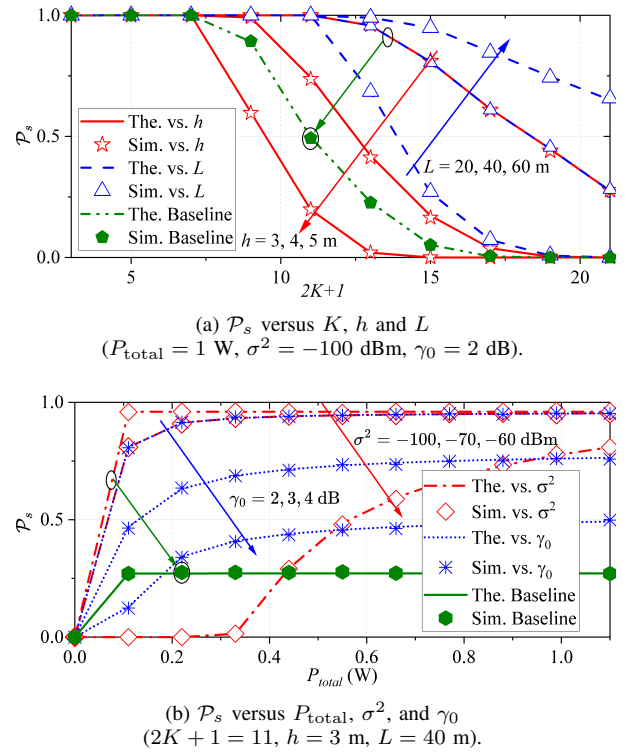


Fig. 4: P_s versus K , h , L , D , γ_0 , P_{total} , and σ^2 ($\mathbf{u}_i : (3, 0, 0)$).

creases, both the per-waveguide power allocation and the inter-waveguide spacing decrease, while the number of interference sources grows. The latter effect reduces the SINR at the user, leading to a decrease in P_s .

- P_s decreases as h increases or L increases. It occurs because a larger h extends the propagation distance and lowers the received power for both desired and interference signals. Based on our setting, the power reduction of the desired signal dominates over that of the interference, leading to a final decrease in SINR and P_s . Meanwhile, a larger L disperses interfering PAs, reducing the aggregated interference and increasing P_s .
- P_s increases with the increasing P_{total} , the decreasing σ^2 , or the decreasing γ_0 . First, the larger P_{total} or the smaller σ^2 leads to the larger P_t (Refer to the SINR model in Eq. (4)), thus enhancing the received SINR at u_i and resulting in a higher P_s . Then, a larger γ_0 raises the decoding threshold, thereby lowering P_s .
- As shown by the green curves in Fig. 4a (i.e., $\gamma_0=2$ dB, $\sigma^2=-100$ dBm) and Fig. 4b (i.e., $h=3$ m, $L=40$ m), PASS consistently outperforms the conventional fixed-position antenna system. This performance gain stems from the ability of PAs to flexibly position themselves closest to the user $\mathbf{u}_i$, thereby enhancing signal reception quality.

Summary: Overall, all simulation results match closely with the theoretical results, thus verifying the accuracy of our model. Meanwhile, the findings from our results can provide many valuable practical insights for network operators and system engineers to optimize parameter configurations in the PASS. For instance, the spatial trends observed in Fig. 3 offer guidance for user deployment strategies aimed at maximizing the downlink successful transmission probability across dif-

ferent waveguide coverage areas. Additionally, the variations in $\mathcal{P}_s$ depicted in Fig. 4 serve as a reference for configuring the PASS under diverse environmental conditions (e.g., room size, network status), enabling power-efficient operation.

V. CONCLUSION

We investigated downlink transmission performance in PA-assisted indoor immersive communications. Specifically, we first developed a 3D spatial model that accurately characterizes the distribution of users, waveguides, and PAs in a typical indoor scenario. We then evaluated the PA-assisted downlink link performance, i.e., STP, which captures the critical location-dependent relationships between system components while quantifying the impacts of key deployment parameters. Extensive Monte Carlo simulations were conducted to verify the accuracy of the proposed theoretical model.

This work has established a formal mathematical foundation for analyzing the network performance of PASS under ideal channel and user distribution assumptions. Future research will extend this foundation to address more practical and complex scenarios, including (i) multi-room environments with inter-room interference and non-uniform user distributions, (ii) the integration of multiple access mechanisms to alleviate interference, and (iii) the mitigation of random blockage effects and measurement of realistic channel characteristics. Moreover, exploring the deployment of multiple PAs per waveguide for enabling beamforming and further enhancing transmission performance represents another promising direction for unlocking the full potential of PASS.

APPENDIX A

Proof of Proposition 1: Substituting Eqs. (2) to (4) into $\mathbb{P}(\gamma_u^i > \gamma_0)$, we have

$$\begin{aligned} \mathcal{P}_s &\stackrel{(a)}{=} \mathbb{P}\left(\frac{((id-y_u^i)^2+h^2)^{-1}}{\eta \sum_{k \in \mathcal{K}/\{i\}} ((x_u^k-x_u^i)^2+(kd-y_u^i)^2+h^2)^{-1} + \frac{\sigma^2}{P_t}} \geq \gamma_0\right) \\ &\stackrel{(b)}{=} \mathbb{P}\left(R \leq \frac{((id-y_u^i)^2+h^2)^{-1}}{\eta \gamma_0} - \frac{\sigma^2}{\eta P_t}\right) \\ &= F_R(z) \Big|_{z=\frac{((id-y_u^i)^2+h^2)^{-1}}{\eta \gamma_0} - \frac{\sigma^2}{\eta P_t}}, \end{aligned} \quad (5)$$

where (a) is resulted from $x_p^i = x_u^i, x_p^k = x_u^k$ (because the PA's location has the same x -axis value with its user) and $y_p^i = id, y_p^k = kd$ (due to the deployment of waveguides) and (b) comes from the substitutions of $R = \sum_{k \in \mathcal{K}/\{i\}} r_k$ and $r_k = ((x_u^k - x_u^i)^2 + (kd - y_u^i)^2 + h^2)^{-1}$. ■

APPENDIX B

Proof of Theorem 1: Let $\phi_R(t)$ denote the characteristic functions of R , the CDF $F_R(z)$ can be obtained by [11]

$$F_R(z) = \frac{1}{2} - \frac{1}{\pi} \int_0^\infty \frac{\text{Im}(e^{-jtz} \phi_R(t))}{t} dt, \quad (6)$$

where $\text{Im}(\cdot)$ calculates the imaginary part. Recall that each PA's location is aligned with its served user and all users are independently distributed. Then the interfering PAs $\{p_k, k \in$

$\mathcal{K}/\{i\}$ are independently and uniformly distributed. Let $\phi_{r_k}(t)$ denote the characteristic functions of r_k , we have

$$\phi_R(t) = \phi_{\sum_{k \in \mathcal{K}/\{i\}} r_k}(t) \stackrel{(a)}{=} \prod_{k \in \mathcal{K}/\{i\}} \phi_{r_k}(t), \quad (7)$$

$$\phi_{r_k}(t) = \mathbb{E}_{r_k} [e^{jtr_k}] \stackrel{(a)}{=} \mathbb{E}_{x_u^k} [e^{jtr_k}] \stackrel{(b)}{=} \frac{1}{L} \int_{-L/2}^{L/2} e^{jtr_k} dx_u^k, \quad (8)$$

where Eq. (7)(a) comes from the independent characteristic of $\{r_k, k \in \mathcal{K}/\{i\}\}$. In Eq. (8), (a) comes from the variable x_u^k in r_k , (b) is resulted from the uniform distribution of $x_u^k \in [-L/2, L/2]$ in the service region $\mathcal{A}_k$ of the waveguide w_k , where probability density function (PDF) of x_u^k is $f_{x_u^k}(x_u^k) = 1/L$, and (c) is due to the identical distribution of $\{x_u^k, k \in \mathcal{K}/\{i\}\}$. Substituting Eqs. (6) to (8) into Eq. (5), we have

$$\begin{aligned} \mathcal{P}_s &= \frac{1}{2} - \frac{1}{\pi} \int_0^\infty \text{Im} \left(e^{-jtz} \prod_{k \in \mathcal{K}/\{i\}} \phi_{r_k}(t) \right) \frac{dt}{t} \\ &\stackrel{(a)}{=} \frac{1}{2} - \frac{1}{\pi} \int_0^\infty \text{Im} \left(e^{-jtz} \left(\frac{1}{L} \right)^{2K} \underbrace{\int_{-L/2}^{L/2} \dots \int_{-L/2}^{L/2} e^{jt \sum_{k \in \mathcal{K}/\{i\}} r_k} \underbrace{dx_u^k \dots dx_u^k}_{k \in \mathcal{K}/\{i\}}}_{2K} \right) \frac{dt}{t} \\ &\stackrel{(b)}{=} \frac{1}{2} - \frac{1}{\pi} \left(\frac{1}{L} \right)^{2K} \int_0^\infty \text{Im} \left(\underbrace{\int_{-L/2}^{L/2} \dots \int_{-L/2}^{L/2} e^{jtA} \underbrace{dx_u^k \dots dx_u^k}_{k \in \mathcal{K}/\{i\}}}_{2K} \right) \frac{dt}{t}, \end{aligned}$$

where (a) is the independent distribution of x_u^k and (b) comes from the substitution of $A = -z + \sum_{k \in \mathcal{K}/\{i\}} (r_k)$. ■

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